

Instrument Variables for Pricing Causal Inference

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1. Executive cheat sheet

Suppose we want the causal effect of a treatment D on an outcome Y , but D is confounded. For pricing, D might be an offer price or a listing-price markdown, while Y might be seller acceptance or buyer conversion. A valid instrument Z provides a source of treatment variation that is not contaminated by the confounding that makes ordinary regression unreliable.

Notation used throughout: Z = instrument; D = treatment whose causal effect we want; Y = outcome; X = observed pre-treatment covariates; U = unobserved confounder(s); ϵ = unobserved component of the outcome equation; u = first-stage error term.

Question	What to ask	Pricing translation
Relevance	Does Z actually move D ?	Does assignment or policy materially change offer/list price?
Independence / exogeneity	Is Z independent of unobserved determinants of Y , possibly conditional on X ?	Is assignment as-good-as-random rather than targeted to hard homes/sellers?
Exclusion restriction	Can Z affect Y through any path other than D ?	Does assignment alter messaging, exposure, service, timing, fees, or financing besides price?
Monotonicity (for LATE)	Does Z move treatment in one direction for everyone, with no systematic defiers?	Does an escalation assignment never cause some sellers to receive a lower offer while raising others?

2. What an instrumental variable is

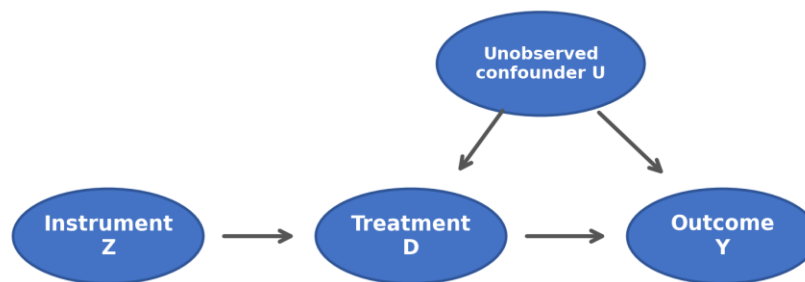
In observational data, treatment is often selected rather than randomly assigned. Price is a classic example: teams lower prices when demand is weak, raise offers when seller acceptance looks unlikely, or change terms in response to home quality and market conditions. Those same factors also affect the outcome, creating endogeneity.

An instrumental variable is a variable Z used to isolate variation in treatment D that can be treated as externally generated. The causal chain we want is:

Desired causal chain

Instrument $Z \rightarrow$ Treatment $D \rightarrow$ Outcome Y

The instrument itself is not the treatment of scientific interest. It is a lever that nudges treatment. We then use only the treatment variation caused by that lever to estimate the causal effect of treatment on outcome.



Valid-IV structure: Z changes D ; Z has no direct path to Y and is independent of U .

Core intuition: use exogenous variation in Z to isolate causal variation in treatment D .

3. Correcting a common misconception about IV

A valid instrument should be independent of the unobserved confounders that jointly affect D and Y . If Z is randomly assigned, or conditionally as-good-as-random, it should not share a common cause with Y that bypasses D . Thus the reduced-form effect of Z on Y is useful causal signal: it is the outcome response generated by the treatment change induced by Z .

Statement	Correct?	Why
" Z must be correlated with Y ."	Not required as a definition	A strong enough causal effect plus relevance will often create a reduced-form association, but sampling noise can make it small.
" Z is correlated with Y because of a confounder."	No, for a valid IV	That would generally violate independence/exogeneity.
" Z causally changes D ."	Yes	This is relevance in causal form.
" Z affects Y only through D ."	Yes	This is the exclusion restriction.
" Z must literally be randomized."	No	True randomization is ideal, but natural or rule-based quasi-random variation can sometimes justify exogeneity.

4. The IV assumptions, one by one

4.1 Relevance

Relevance means the instrument changes or predicts treatment after accounting for included covariates. In a linear first stage:

First-stage equation

$$D = \alpha + \pi \cdot Z + \gamma'X + u.$$

Relevance requires $\pi \neq 0$.

Notation: D = treatment; α (α) = intercept; π (π) = first-stage effect/coefficient of the instrument; Z = instrument; X = vector of observed exogenous controls; γ (γ) = coefficient vector for X ; $\gamma'X$ = the linear contribution of those controls; u = first-stage error term; $\pi \neq 0$ means the instrument moves treatment.

In pricing, a randomized escalation policy is relevant if assignment materially changes the actual offer. A markdown-assignment experiment is relevant if assignment materially changes realized listing price.

4.2 Independence / exogeneity

The instrument must be unrelated to unobserved determinants of the outcome, at least after conditioning on justified covariates. In shorthand, $\text{Cov}(Z, \epsilon) = 0$. Randomization is the cleanest route, but quasi-random assignment, lotteries, thresholds, queues, or operational rules can sometimes provide credible variation.

A common failure is targeted assignment. If difficult-to-convert listings are intentionally sent to a special pricing policy, that policy indicator is not automatically a valid instrument because it is correlated with latent demand.

4.3 Exclusion restriction

Exclusion says Z affects Y only by changing D . It rules out direct effects and alternative mediating channels. A useful thought experiment is: if I could magically hold treatment fixed, would changing Z still change the outcome? If yes, exclusion is questionable.

Candidate Z	Hold price fixed: could Y still change?	Exclusion assessment
Random markdown assignment with no other operational changes	Probably not	Plausible
Local home-price appreciation	Yes: seller expectations and market demand can change	Poor
Mortgage rates	Yes: buyer affordability changes	Poor
Analyst identity	Yes: service, timing, judgment, negotiation can differ	High risk
Sharp pricing threshold	Potentially no, if nothing else changes at the cutoff	Potentially credible / RD-style

4.4 Monotonicity and LATE

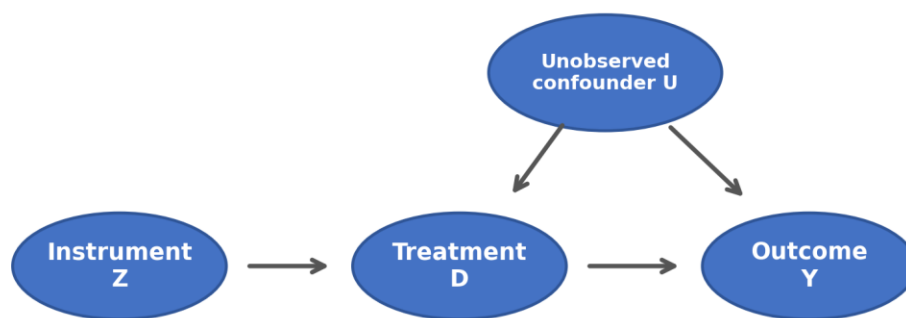
For the classic binary-instrument LATE interpretation, monotonicity means assignment does not push treatment in opposite directions for different people. If Z is an encouragement to raise offers, there should

not be a meaningful subgroup whose offer systematically falls because of the encouragement. Under relevance, independence, exclusion, and monotonicity, IV identifies the Local Average Treatment Effect for compliers: the units whose treatment is actually changed by Z.

4.5 SUTVA (Stable Unit Treatment Value Assumption)/ interference considerations

Standard IV reasoning also relies on stable treatment definitions and limited interference across units. In housing, interference can matter: markdowns on nearby homes may change comparable-sale expectations or buyer traffic for other listings. If one listing's assignment affects another listing's outcome, the estimand and design need extra care.

5. Why IV works: intuition and DAGs



Valid-IV structure: Z changes D; Z has no direct path to Y and is independent of U.

Figure: valid IV structure with an unobserved confounder U of treatment and outcome.

Ordinary regression of Y on D mixes two sources of treatment variation: (1) variation caused by confounders U and (2) externally generated variation caused by Z. IV tries to keep only source (2). Because Z is independent of U and has no path to Y except through D, the Z-induced component of D behaves like quasi-random treatment variation.

Mental model

OLS asks: "When price differs, how does outcome differ?"

IV asks: "When price differs specifically because the instrument moved it, how does outcome differ?"

6. Estimation: Wald estimator and 2SLS

6.1 Binary instrument: Wald estimator

With a binary instrument Z and a scalar treatment D, a simple IV estimand is:

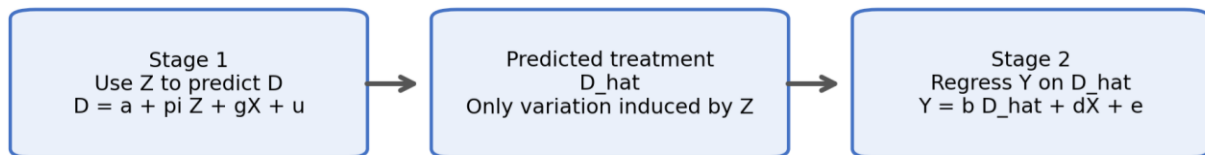
Wald ratio

IV effect = $[E(Y | Z=1) - E(Y | Z=0)] / [E(D | Z=1) - E(D | Z=0)]$.

Notation: $E(\cdot)$ = expectation/average; Y = outcome; D = treatment; Z = binary instrument; $Z=1$ = assigned/encouraged group; $Z=0$ = control group. The numerator is the reduced-form effect of Z on Y ; the denominator is the first-stage effect of Z on D ; their ratio is the IV/Wald estimand under the IV assumptions.

The numerator is the reduced-form effect of assignment on the outcome. The denominator is the first-stage effect of assignment on treatment. Dividing tells us how much outcome changes per unit of instrument-induced treatment change.

6.2 Two-stage least squares (2SLS)



2SLS isolates the part of treatment variation generated by the instrument.

- Stage 1: predict treatment using the instrument and justified controls. Save $D_{\hat{}}$.
- Stage 2: regress outcome on $D_{\hat{}}$ and the same justified controls.
- Use IV/2SLS software for inference rather than manually running two OLS regressions, because standard errors must account for the generated regressor and design.

For nonlinear binary outcomes, a linear probability 2SLS model is often used for a transparent local effect, but more specialized IV estimators may be appropriate depending on the estimand, treatment, and outcome. Do not casually run a first-stage logistic regression and plug predictions into a second-stage logistic regression and call it standard 2SLS.

7. LATE, compliers, and what IV actually identifies

IV frequently does not identify the average treatment effect for every listing or seller. With a binary instrument and heterogeneous treatment effects, the classic result is a Local Average Treatment Effect (LATE) among compliers.

Type	Behavior when Z switches 0 -> 1	Role
Complier	Takes the treatment because of the instrument	IV identifies the effect for this group under assumptions
Always-taker	Treated regardless of Z	Does not contribute to first-stage contrast
Never-taker	Untreated regardless of Z	Does not contribute to first-stage contrast
Defier	Moves opposite to the encouragement	Monotonicity rules out systematic defiers

Pricing translation: if markdown assignment only changes actual price for a subset of listings because humans can override it, the estimated effect pertains to listings whose price is changed by the assignment. That group may differ from all listings, so external validity must be discussed explicitly.

8. Weak instruments and first-stage diagnostics

A weak instrument barely moves treatment. Weak IV can produce unstable estimates, large finite-sample bias, and misleading inference. The classic first-stage F-statistic is a screening diagnostic in simple linear settings, but the old " $F > 10$ " rule is only a rough heuristic and can be inadequate with multiple instruments, heteroskedasticity, clustering, or weak-identification concerns.

Diagnostic	What it tells you	What it does NOT tell you
First-stage coefficient	Direction and size of $Z \rightarrow D$	Whether exclusion is valid
First-stage F / weak-IV statistics	How much identifying strength Z provides	Whether Z is exogenous
Balance tests under randomization	Whether observed covariates look balanced	Whether unobserved balance or exclusion is proven
Reduced form $Z \rightarrow Y$	Whether assignment changes the outcome	Which causal channel produced the change
Overidentification tests with multiple IVs	Whether instruments are mutually compatible with restrictions under model assumptions	A definitive proof that every instrument is valid

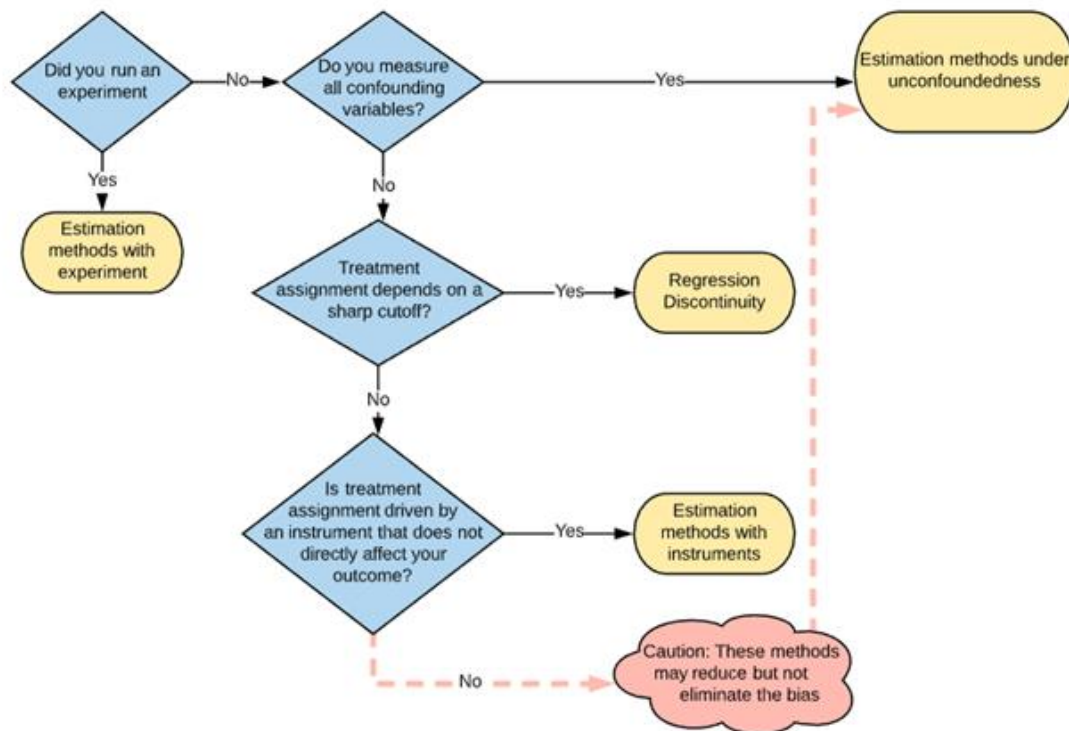
9. IV versus alternative causal designs

RCT is the traditional gold standard that enables unbiased estimation of treatment effects. However, in observational studies, the possibility of bias arises because a difference in the treatment outcome may be caused by a factor that predicts treatment rather than the treatment itself, known as confoundedness. As a result, it's desirable to replicate a randomized experiment as closely as possible through various strategies. In general, Causal inference consists of a family of statistical methods including estimation methods under unconfoundedness (also known as conditioning-based methods), and estimation methods for quasi-experiments (a research design that looks like an experimental design but lacks the key ingredient — random assignment of the treatment — studying instead pre-existing groups that received different treatments after the fact). In quasi-experiments, commonly used approaches include simple natural experiments, instrumental-variables (IV), and regression-discontinuity models. The following shows high-level workflow recommendations for causal algorithm selection for an observational study with one snapshot of the outcome.

A few example methods are listed below. These methods are complementary rather than mutually exclusive. For example, a randomized encouragement design can be analyzed using instrumental variables (IV), since assignment is randomized while treatment compliance may be imperfect. This is not intended to be a comprehensive list, but rather a set of illustrative examples.

Method	Main identification idea	Best fit	Main weakness in pricing
Randomized controlled trial	Randomize treatment D itself	You can directly randomize offer/list price or policy	Operational, legal, fairness, or revenue constraints
Instrumental variables	Randomize or quasi-randomize Z that shifts endogenous D	Treatment cannot be cleanly randomized but an external lever exists	Exclusion and exogeneity are hard to defend; often local effect

Difference-in-differences	Compare treated vs control changes around a policy	Market/policy rollout with credible parallel trends	Parallel trends can fail; concurrent shocks
Regression discontinuity	Exploit a treatment jump at a cutoff	Sharp/fuzzy pricing or eligibility threshold	Usually local to cutoff; manipulation or co-jumps can invalidate
Structural demand/pricing model	Specify economic behavior and estimate primitives	Optimization, counterfactual pricing, policy simulation	More modeling assumptions; misspecification risk



Please keep in mind that the list is not exhaustive, just a small representation of the approaches that I used on the Customer Growth Analytics team here at Microsoft.

10. Opendoor-style IV candidate library

The following examples are deliberately framed as interview thought experiments. They illustrate how to reason about IV validity; they do not imply these mechanisms exist in production.

Candidate instrument	Side	Treatment D	Outcome Y	Relevance	Exclusion risk	How I would frame it
Randomized offer-escalation assignment	Buy side	Actual offer	Seller acceptance	Strong by design	Low if only offer changes	Preferred hypothetical IV
Randomized markdown assignment	Sell side	Actual list price	Buyer conversion / sale	Strong by design	Low if exposure and terms unchanged	Preferred hypothetical IV

Randomized pricing recommendation shown to human	Either	Implemented price	Acceptance / conversion	Depends on compliance	Low only if recommendation affects nothing else	Good encouragement design
Queue-based random reviewer assignment	Buy side	Offer / adjustment	Acceptance	May move price	Reviewer may change service/timing	Risky without operational isolation
Analyst / pricer identity	Either	Price	Conversion	May be strong	High: judgment, negotiation, timing	Usually risky
Recent local appreciation	Either	Price / valuation	Acceptance / conversion	Often strong	High: seller beliefs, outside option, demand	Invalid teaching example
Mortgage-rate movement	Sell side	Price / markdown	Buyer conversion	Can be strong	High: affordability directly changes	Usually poor
Weather shock	Sell side	Price response	Tours / conversion	May move pricing	Directly changes search/tours	Poor
Automated pricing threshold	Sell side	Markdown	Conversion	Potentially sharp	Credible if only treatment jumps	Consider fuzzy RD
Eligibility threshold for escalation	Buy side	Offer escalation	Acceptance	Potentially sharp	Credible if no other policy changes at cutoff	Consider fuzzy RD
Model version rollout by time	Either	Price	Conversion	Can move price	Time trends and simultaneous launches	Risky; maybe DiD/event study
Geographic rollout of new pricing policy	Either	Price	Conversion	Can move price	Markets differ; local shocks	Often DiD, not automatically IV
System outage affecting recommendation adoption	Either	Implemented price	Conversion	May create compliance shock	Outage may affect latency/service too	Potential natural experiment, high scrutiny
Randomized seller email reminder	Buy side	Negotiation / action	Acceptance	Could affect engagement	Directly affects acceptance	Not an IV for price; treatment itself may be messaging
Randomized ad boost	Sell side	Traffic / exposure	Conversion	Strong for exposure	Directly changes conversion through exposure	Could instrument exposure, not price
Randomized financing incentive	Sell side	Effective monthly cost	Conversion	Strong	May change product attractiveness directly	Define treatment carefully; exclusion depends on estimand

10.1 Preferred buy-side example: randomized escalation

Allowed path



Exclusion requires: assignment does NOT also change messaging, service level, speed, fees, or contract terms.

Hypothetical buy-side encouragement design.

Suppose eligible seller leads are randomly assigned to an escalation policy. Assignment changes the probability or amount of offer increase, but seller-facing messaging, SLA, inspection policy, fees, and closing terms are held constant. Assignment is then a credible instrument for actual offer amount. If humans can override the recommendation, IV naturally handles noncompliance and estimates a local effect for cases whose offer is changed by assignment.

10.2 Preferred sell-side example: randomized markdown assignment



Exclusion requires: assignment does NOT simultaneously create a "price drop" badge, ad boost, extra emails, or other demand shock.

Hypothetical sell-side randomized markdown instrument.

The design is clean only if the random assignment alters price and does not simultaneously alter marketing treatment. A "price reduced" badge, email campaign, search-ranking boost, open-house intensity, or agent follow-up would create additional paths from Z to Y and violate exclusion for the causal effect of price.

10.3 Why recent local appreciation is a bad IV



Why it fails exclusion: appreciation can change seller expectations and market demand even if Opendoor price were held fixed.

Recent local appreciation has multiple plausible paths to the outcome besides Opendoor price.

Recent local home-price appreciation means the percentage increase in comparable local home values over a recent window such as 3, 6, or 12 months. It may strongly predict an algorithmic valuation or offer. But it can also increase a seller's reservation value, change expectations about waiting to sell, and alter buyer demand. Therefore it can be highly relevant and still fail exclusion.

10.4 Mortgage rates

Mortgage rates can move prices, markdown decisions, and market conditions, so relevance may be substantial. But they also directly change buyer affordability and monthly payment. On the seller side, rates can affect the attractiveness of moving and obtaining a replacement mortgage. Thus holding price fixed does not hold the outcome fixed when rates change: a classic exclusion concern.

10.5 Analyst or pricer identity

Analyst identity sometimes looks attractive because different analysts may systematically recommend different prices. The problem is that identity can also alter response time, negotiation style, escalation behavior, comp selection, exception handling, communication quality, or case mix. Random assignment plus strict workflow standardization would improve the design, but identity is not automatically a valid instrument merely because assignment looks operational.

10.6 Pricing thresholds and fuzzy RD

Suppose a score of 0.500 triggers a recommended 3% markdown while 0.499 does not. Listings just around the threshold may be comparable. If actual markdown compliance is imperfect, threshold crossing can serve as an instrument for treatment in a fuzzy regression-discontinuity design. The exclusion logic becomes: nothing else important should jump at the same threshold, and units should not precisely manipulate the running variable around the cutoff.

11. From causal price elasticity to pricing optimization

We often care less about the causal coefficient itself than about how it enters a pricing decision. The key distinction is prediction versus intervention. A predictive model estimates $P(Y=1 \mid \text{price}, X)$. A causal pricing model needs $P(Y=1 \mid \text{do}(\text{price}), X)$, or a defensible approximation to how conversion would change if we actively changed price.

11.1 Sell-side objective

Illustrative one-horizon objective

Expected contribution(p) = $q(p) * [p - \text{basis} - \text{variable_cost}] - [1 - q(p)] * \text{expected_non_sale_cost}$.

Here $q(p)$ is the causal probability of sale within a decision horizon at price p . The non-sale cost can represent expected carrying cost, future markdown risk, financing cost, inventory capital, or an approximation to continuation value. A production system would typically use a richer dynamic objective rather than this one-period simplification.

If a local IV estimate implies that a 1% price reduction raises 30-day conversion by about 2 percentage points near the current price, the optimizer can compare the loss in per-sale margin against the gain in sale probability and reduced inventory cost. The correct decision depends on basis, holding cost, time-to-sale value, uncertainty, and whether the local slope remains valid for the contemplated change.

11.2 Buy-side objective

Illustrative buy-side objective

Expected lead value(offer) = $a(\text{offer}) * [\text{expected_resale_net_value} - \text{offer} - \text{transaction_costs}]$, where $a(\text{offer})$ is causal seller acceptance probability.

Offer	Illustrative acceptance	Margin if acquired	Expected lead value
\$480k	25%	\$50k	\$12.5k
\$490k	35%	\$40k	\$14.0k
\$500k	45%	\$30k	\$13.5k

The highest offer is not automatically optimal: acceptance rises with offer, but unit economics fall. Causal estimation matters because using a confounded acceptance model can overstate or understate the incremental acceptance generated by another \$1,000 of offer.

11.3 Why IV alone may not be enough for global optimization

- A single binary instrument often gives one local slope or one LATE, not a full response curve.
- Optimization across a wide price range requires richer variation: multiple randomized price doses, continuous encouragement, repeated experiments, or a structural demand model.
- Effects can vary by home liquidity, market tightness, price band, seller urgency, season, days on market, and distance from comparable sales.
- Uncertainty should enter guardrails: constrain extrapolation outside support and shrink noisy segment-level effects.
- Long-run effects matter: a markdown can change days on market, future buyer beliefs, negotiation, and carrying cost, not just immediate conversion.

12. Heterogeneity, nonlinear effects, and multiple price doses

A practical pricing system should expect heterogeneous treatment effects. A 1% markdown on a highly liquid starter home may have a smaller conversion effect than the same percentage change on a stale, mispriced luxury listing. Likewise, a \$5,000 offer increase may matter more for a seller near an acceptance threshold than for a seller whose outside option is far above the offer.

Approach	What it adds	Caution
Multiple randomized price arms	Estimates response at several price changes	Requires enough sample and business tolerance
Continuous randomized encouragement	Creates richer treatment variation	Need monotonic, interpretable encouragement mechanism

Interactions in 2SLS	Allows effects by observable segment	Weak first stages within segments can explode variance
Causal forests / heterogeneous IV methods	Flexible heterogeneity discovery	Interpretability and overlap; avoid data dredging
Structural demand model	Full counterfactual demand/optimization framework	Strong behavioral assumptions and calibration burden

13. Diagnostics, falsification, and robustness checks

No statistical test can prove exclusion, because exclusion is fundamentally a causal assumption about pathways. Good IV work therefore combines design knowledge, institutional detail, falsification tests, and sensitivity analysis.

- First stage: report effect size, uncertainty, and weak-IV diagnostics, not just significance.
- Randomization/balance: verify instrument assignment is balanced on pre-treatment home, seller, market, and operational covariates.
- Pre-treatment outcomes: Z should not predict outcomes measured before assignment.
- Operational invariance: check that Z does not change response time, contact frequency, channel, agent seniority, marketing exposure, financing, inspection treatment, or fees unless those are part of D.
- Placebo thresholds or placebo dates: useful in RD or rollout designs.
- Density/manipulation checks near a threshold: look for sorting around the cutoff.
- Reduced form: report $Z \rightarrow Y$ alongside first stage and IV estimate.
- Alternative windows/specifications: verify conclusions are not driven by one bandwidth, time window, or functional form.
- Clustered inference: account for geographic, temporal, seller, or listing-level dependence where relevant.
- Weak-IV robust inference: consider Anderson-Rubin or related procedures when identification may be weak.
- External validity: describe who the compliers are and where the instrument actually moves price.

14. Common IV failure modes

Failure mode	Example	Why it biases IV
Direct effect of Z on Y	Markdown assignment also triggers marketing emails	Outcome changes through marketing as well as price
Z shares a cause with Y	Analysts assigned difficult cases	Assignment correlates with latent conversion difficulty
Bad control / post-treatment control	Control for a mediator affected by Z	Can block part of causal pathway or induce collider bias
Weak instrument	Recommendation barely changes actual price	Estimator becomes unstable and weak-ID bias grows
Multiple mechanisms bundled in instrument	Escalation policy changes price, fees, and closing speed	Cannot attribute reduced form to price alone
Time confounding	New model version launched during	Z captures time trend and market

	seasonal demand shift	shock
Geographic confounding	Policy only in fast-growing markets	Market differences affect outcome directly
Defiers / non-monotone response	Encouragement raises price for some but lowers it for others	Classic LATE interpretation may fail
Interference	One home markdown changes traffic or comps for nearby homes	One unit's Z affects another unit's outcome
Extrapolation	Use local IV slope for a 15% price change	Local estimand may not describe distant counterfactual

15. Glossary and references

15.1 Glossary

Term	Meaning
Endogeneity	Treatment is correlated with unobserved determinants of the outcome, often because of confounding, simultaneity, or selection.
Instrument Z	A variable used to isolate exogenous variation in treatment D.
First stage	Relationship from instrument Z to treatment D.
Reduced form	Relationship from instrument Z to outcome Y.
Exclusion restriction	Z affects Y only through D.
Exogeneity / independence	Z is independent of unobserved outcome determinants, perhaps conditional on justified covariates.
LATE	Local Average Treatment Effect for compliers whose treatment is changed by Z.
Complier	A unit whose treatment responds in the intended direction to the instrument.
Weak instrument	An instrument that provides little variation in treatment.
2SLS	Two-stage least squares, the standard linear IV estimator.
Fuzzy RD	Regression discontinuity where crossing the cutoff changes treatment probability rather than deterministically setting treatment.
Reservation value	The minimum value/price at which a seller is willing to transact, given outside options and beliefs.

15.2 Core references for further study

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